

Day 2: Network Topologies

1. What is Network Topology?

Network topology means **how computers and network devices are connected to each other**.

It shows:

- How devices are connected
- How data travels
- What happens if one connection fails
- How the network can be expanded
- Which device is the central point

Simple Example

In a computer lab:

```
PC1
|
PC2 ---- Switch ---- PC3
|
PC4
```

The way these devices are connected is called the **network topology**.

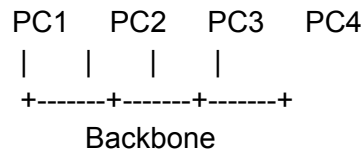
Today we will learn:

1. Bus Topology
 2. Star Topology
 3. Ring Topology
 4. Mesh Topology
-

2. Bus Topology

What is Bus Topology?

In a bus topology, all computers are connected to a **single main cable**, called the backbone cable.



All devices share the same main communication path.

How Does It Work?

Suppose PC1 wants to send data to PC3.

PC1 → Main Cable → PC3

The data travels through the common cable.

Other computers can see the transmission, but the intended device processes the data.

Advantages

- Simple design
- Requires less cable
- Low cost for a small setup
- Easy to understand

Disadvantages

- If the main cable fails, the network can be affected.
- Performance decreases as more devices are added.
- Troubleshooting can be difficult.
- Limited scalability.
- Collisions can occur in traditional shared Ethernet implementations.

Real-Life Example

Older small computer networks used bus-style network arrangements.

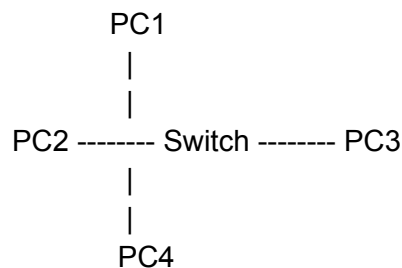
Today, bus topology is mainly useful for **learning and understanding networking concepts** rather than building modern LANs.

3. Star Topology

What is Star Topology?

In a star topology, all devices connect to a **central device**.

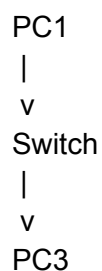
The central device is commonly a **switch** in a modern Ethernet LAN.



Each computer has a separate connection to the switch.

How Does It Work?

Suppose PC1 wants to send data to PC3:



The switch receives the frame and forwards it toward the appropriate device.

Advantages

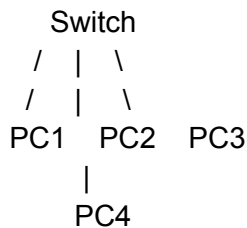
- Easy to install
- Easy to troubleshoot
- Easy to add or remove devices
- Failure of one cable usually affects only that device.
- Good performance with switches
- Commonly used in modern LANs

Disadvantages

- Requires more cable than bus topology.
 - Requires a central device.
 - If the central switch fails, connected devices lose LAN connectivity through that switch.
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Real-Life Example

A college computer lab:

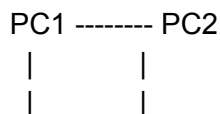


This is a typical **star topology**.

4. Ring Topology

What is Ring Topology?

In a ring topology, each device is connected to two other devices, forming a closed loop.



PC4 ----- PC3

The devices form a ring.

How Does It Work?

A simplified example:

PC1 → PC2 → PC3 → PC4 → PC1

Data can travel around the ring according to the technology and design.

Advantages

- Organized connection structure
- Each device has a defined connection path.
- Can provide predictable access in some ring-based technologies.

Disadvantages

- A connection failure can disrupt communication in a simple single ring.
- Adding or removing devices can be more complicated.
- Troubleshooting can be difficult.
- Less common for ordinary modern Ethernet LANs.

Some technologies use **dual rings or other protection mechanisms** to improve resilience.

Real-Life Example

Ring structures have historically been used in some telecommunications and network technologies.

For classroom understanding, imagine computers arranged in a circle where each computer connects to the next one.

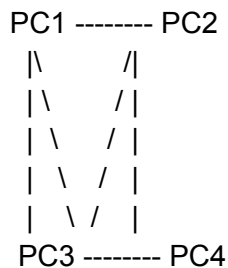
5. Mesh Topology

What is Mesh Topology?

In a mesh topology, devices have multiple connections with other devices.

In a **full mesh**, every device is directly connected to every other device.

Example with four devices:



The exact drawing can vary, but the important point is that there are **multiple paths** between devices.

Full Mesh Example

For 4 devices:

PC1 ↔ PC2
PC1 ↔ PC3
PC1 ↔ PC4
PC2 ↔ PC3
PC2 ↔ PC4
PC3 ↔ PC4

Every device has a direct connection to every other device.

Advantages

- Multiple paths are available.
- High redundancy.
- If one connection fails, another path may still be available.

- Suitable where reliability is very important.

Disadvantages

- Requires many connections.
 - Expensive.
 - Installation is complex.
 - Maintenance is difficult as the number of devices increases.
-

Real-Life Example

Mesh concepts are useful in environments where **redundancy and availability** are important.

For example, network infrastructure may use multiple links between important devices so that traffic can take another path if one link fails.

6. Topology Comparison

Topology	Main Idea	Central Device	Reliability	Cost	Common Use
Bus	One shared backbone cable	No	Low	Low	Historical/small networks
Star	All devices connect to center	Yes	Good	Medium	Modern LANs
Ring	Devices form a loop	No	Medium	Medium	Some specialized networks
Mesh	Multiple connections/paths	No single central device	High	High	High-redundancy networks

7. Easy Way to Remember

Bus

One main road

All devices use the same road.

Star

One central point

All devices connect to a central switch.

Ring

Circle

Devices form a closed loop.

Mesh

Many paths

Devices have multiple connections.

8. Real-Life Comparison

Imagine a college.

Bus

One common cable

If the main cable has a problem, many devices can be affected.

Star

Every classroom PC → Switch

If one PC cable fails, other PCs can continue working.

Ring

PC1 → PC2 → PC3 → PC4 → PC1

Devices form a closed loop.

Mesh

Multiple connections between important network devices

If one path fails, another path may be available.

9. Practical – Cisco Packet Tracer

Objective

Create and test a **Star Topology** using Cisco Packet Tracer.

Required Devices

Add:

- 1 Switch
 - 4 PCs
 - Copper Straight-Through cables
-

10. Step 1 – Open Cisco Packet Tracer

Open:

Cisco Packet Tracer

You will see the workspace.

At the bottom, you will find different categories of networking devices.

11. Step 2 – Add a Switch

From the device section:

Network Devices



Switches

Select a basic switch such as:

2960

Drag it into the workspace.

12. Step 3 – Add Four PCs

Go to:

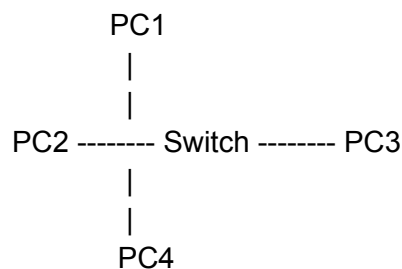
End Devices

Select:

PC

Add four PCs.

Arrange them like this:



13. Step 4 – Connect the PCs

Select the connection/cable tool.

Choose:

Copper Straight-Through

Connect:

PC1 → Switch
PC2 → Switch
PC3 → Switch
PC4 → Switch

For example:

PC1 FastEthernet0 → Switch FastEthernet0/1

PC2 FastEthernet0 → Switch FastEthernet0/2

PC3 FastEthernet0 → Switch FastEthernet0/3

PC4 FastEthernet0 → Switch FastEthernet0/4

The exact available port numbers may differ depending on the device selected.

14. Step 5 – Configure IP Addresses

We need IP addresses so the PCs can communicate.

Use:

Device	IP Address	Subnet Mask
PC1	192.168.10.1 0	255.255.255. 0
PC2	192.168.10.2 0	255.255.255. 0
PC3	192.168.10.3 0	255.255.255. 0
PC4	192.168.10.4 0	255.255.255. 0

No default gateway is required for this same-LAN demonstration.

15. Configure PC1

Click:

PC1

Go to:

Desktop



IP Configuration

Enter:

IP Address:

192.168.10.10

Subnet Mask:

255.255.255.0

Do the same for the other PCs.

16. Configure PC2

IP Address:

192.168.10.20

Subnet Mask:

255.255.255.0

17. Configure PC3

IP Address:

192.168.10.30

Subnet Mask:

255.255.255.0

18. Configure PC4

IP Address:
192.168.10.40

Subnet Mask:
255.255.255.0

19. Step 6 – Test Connectivity

Open PC1:

PC1



Desktop



Command Prompt

Run:

ping 192.168.10.20

This tests PC1 → PC2.

Then:

ping 192.168.10.30

And:

ping 192.168.10.40

If the configuration is correct, the PCs should be able to communicate through the switch.

20. Important Practical Observation

Ask students:

What happens if the cable between PC1 and the switch is removed?

Expected concept:

PC1 ✗ Switch

PC1 loses its connection to the LAN.

But:

PC2 ↔ Switch ↔ PC3

PC4 ↔ Switch

can continue communicating, assuming their connections remain active.

This demonstrates an important advantage of star topology.

21. Practical Experiment – Switch Failure

Now ask:

What happens if the central switch fails?

The result is:

PC1 ✗

PC2 ✗

PC3 ✗

PC4 ✗

The connected PCs can no longer communicate through that switch.

This demonstrates the importance of the **central device** in a star topology.

22. Simulation Mode in Packet Tracer

Cisco Packet Tracer also has:

Realtime
Simulation

Choose:

Simulation

Then generate a ping between two PCs.

For example:

PC1 → ping → PC3

Observe how the packet travels:

```
PC1
↓
Switch
↓
PC3
```

Students should observe that the switch is the central connection point.

23. Practical Activity – Compare Topologies

After creating the star topology, ask students to draw the other topologies.

Bus

```
PC1  PC2  PC3  PC4
|    |    |    |
+----+----+----+
Main Cable
```

Star

```
    PC1
    |
PC2 --- Switch --- PC3
    |
    PC4
```

Ring

```
PC1 ---- PC2
|        |
```

| |
PC4 ----- PC3

Mesh

PC1 ----- PC2
|\ /|
|\ /|
|\ /|
|\ /|
|\ /|
PC3 ----- PC4

24. Task – Topology Diagram

Students have to create a document containing all four topologies.

For each topology, include:

1. Name

Example:

Star Topology

2. Diagram

Draw the topology.

3. How it works

Write 2–3 simple sentences.

4. Advantages

Write at least 2 points.

5. Disadvantages

Write at least 2 points.

6. Real-Life Example

Give one practical example.

25. Student Assignment Format

Students can prepare this table:

Topology	Diagram	Main Feature	Advantages	Disadvantages	Example
Bus	Draw	Single backbone	Low cost, simple	Backbone failure affects network	Older LAN
Star	Draw	Central switch	Easy troubleshooting	Central device failure affects connected devices	Computer lab
Ring	Draw	Closed loop	Organized path	Failure can affect simple ring	Specialized networks
Mesh	Draw	Multiple paths	High redundancy	Expensive and complex	High-availability network

26. Final Practical Challenge

Give students this requirement:

A college has 5 computers in one computer lab. Create a suitable topology in Cisco Packet Tracer and test communication between all computers.

Students should:

1. Add one switch.
2. Add five PCs.
3. Connect all PCs to the switch.
4. Assign IP addresses.
5. Use the same subnet.
6. Test PC-to-PC communication using **ping**.
7. Use Simulation Mode.

8. Remove one PC cable and test again.
9. Explain what happened.
10. Draw the final topology diagram.

Suggested IPs

PC1 → 192.168.10.10

PC2 → 192.168.10.20

PC3 → 192.168.10.30

PC4 → 192.168.10.40

PC5 → 192.168.10.50

Subnet Mask:
255.255.255.0

Day 2 Learning Outcome

After this session, students should be able to:

- Explain what network topology means.
- Identify Bus, Star, Ring, and Mesh topology.
- Explain the working of each topology.
- Compare advantages and disadvantages.
- Identify suitable real-world examples.
- Create a star topology in Cisco Packet Tracer.
- Configure basic IP addresses in Packet Tracer.
- Test PC-to-PC connectivity using **ping**.
- Observe packet movement using Simulation Mode.
- Understand the effect of cable and central-device failure.
- Draw and document different network topologies.